Answer key: Lesson 10, Story Steps: Read It Twice

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Ben has 30 cards. He gives 20 cards away. After that, he gets 6 more cards.

Answer: Step 1: 30 - 20 = 10. Step 2: 10 + 6 = 16. Ben has 16 cards.

First, Ben gives away 20, so we take away: 30 - 20 = 10. Then he gets 6 more, so we add: 10 + 6 = 16. Ben has 16 cards now.

Watch for: Adding the 20 (30 + 20) because 20 is in the story.; Using 30 again in Step 2 instead of the 10 left after Step 1.

S2. Mei has 42 stickers. She gets 7 more stickers. Then she gives 10 stickers to Ali. Write one number sentence for each step.

Answer: Step 1: 42 + 7 = 49. Step 2: 49 - 10 = 39. Mei has 39 stickers.

Step 1: she gets 7 more, so add: 42 + 7 = 49. Step 2: she gives 10 away, so take away: 49 - 10 = 39. Mei has 39 stickers now.

Watch for: Using 42 again in Step 2 (42 - 10 = 32).; Adding both numbers (42 + 7 + 10).

S3. Ravi has 25 pencils. He buys 10 more pencils. Then he loses 3 pencils. Circle the 2 number sentences that fit the story.

Answer: Circle 25 + 10 = 35 and 35 - 3 = 32. Ravi has 32 pencils.

Step 1: buys 10 more, so 25 + 10 = 35. Step 2: loses 3, so 35 - 3 = 32. The sentences 25 - 10 = 15 and 35 + 3 = 38 use the wrong sign.

Watch for: Circling 25 - 3 = 22 because both numbers are in the story.; Circling a sentence with the wrong sign.

S4. Write + or - in each box.

Answer: +, -, -, +

Finds more means add (+). Eats means take away (-). Gives away means take away (-). Join means more, so add (+).

Watch for: Choosing + for gives away because there is a number in the sentence.; Choosing - for join.

S5. Ali has 23 marbles. Ben has 10 more than Ali. How many marbles do they have altogether? Use counters if you like.

Answer: Step 1: Ben has 23 + 10 = 33. Step 2: 23 + 33 = 56. They have 56 marbles.

Step 1: Ben has 10 more than Ali, so 23 + 10 = 33. Step 2: altogether means add Ali and Ben: 23 + 33 = 56 marbles.

Watch for: Answering 33 (stopping after Step 1).; Adding 23 + 10 and calling that altogether.

S6. A bus has 30 children. 10 children get off. Then 4 children get on. Siti writes 30 + 10 = 40 for Step 1.

Answer: No. Get off means take away: 30 - 10 = 20. Then 20 + 4 = 24. There are 24 children.

Children who get off leave the bus, so Step 1 is a take away: 30 - 10 = 20. Siti added the 10. Then 4 get on, so add: 20 + 4 = 24.

Watch for: Circling Yes.; Adding both numbers to 30 (30 + 10 + 4 = 44).

Think harder (page 3)

T1. Ravi has 15 marbles. He has 3 more marbles than Mei.

Answer: Mei has 12 marbles. Together they have 27.

Ravi has 3 more than Mei, so Mei has 3 fewer than Ravi: 15 - 3 = 12. Together: 15 + 12 = 27 marbles.

Watch for: Adding 15 + 3 = 18 because the word more is there.; Answering 12 for the second part.

T2. Mei is in a line to buy lunch. There are 6 children in front of her and 4 children behind her. How many children are in the line?

Answer: 11 children (6 + 4 + 1, counting Mei).

The children in front are 6 and behind are 4: 6 + 4 = 10. Mei is in the line as well, so 10 + 1 = 11 children.

Watch for: Answering 10 and leaving Mei out.

T3. Ben has 12 marbles. Mei has 5 more marbles than Ben. Siti has 3 fewer marbles than Mei. How many marbles does Siti have?

Answer: Mei has 17. Siti has 14 marbles.

Step 1: Mei has 5 more than Ben: 12 + 5 = 17. Step 2: Siti has 3 fewer than Mei: 17 - 3 = 14 marbles.

Watch for: Taking 3 from 12 (Siti is compared with Mei, not Ben).; Answering 17 (stopping after Step 1).

T4. Ali has 12 stickers. He gives 7 stickers to Mei. Then he gives 8 stickers to Ben.

Answer: 5 stickers left; No, the story cannot happen.

Step 1: 12 - 7 = 5, so Ali has 5 stickers left. Step 2 needs 8 stickers, but 5 is less than 8. Ali cannot give Ben 8, so this story can not be true. A story step can be impossible, so check each step.

Watch for: Writing 12 - 7 - 8 and getting an answer without noticing 5 is less than 8.; Circling Yes without checking step 2.

Go further (page 4)

G1. A shop has 60 pencils. It sells 24 pencils on Monday and 15 pencils on Tuesday. On Wednesday, a box of 30 pencils comes in. Write one number sentence for each step.

Answer: Step 1: 60 - 24 = 36. Step 2: 36 - 15 = 21. Step 3: 21 + 30 = 51. The shop has 51 pencils.

Monday: 60 - 24 = 36. Tuesday: 36 - 15 = 21. Wednesday a box comes in, so add: 21 + 30 = 51 pencils.

Watch for: Adding 24 and 15 and taking that from 60 but forgetting the box.; Starting each step from 60 again.

G2. Step 1: 25 + 10 = 35. Step 2: 35 - 8 = 27. Write a story that fits both steps. End your story with a question. Many stories are right.

Answer: Any story where 25 grows by 10 and then 8 are taken away, and the question asks how many now. The answer is 27.

For example: Ali has 25 stickers. He gets 10 more. Then he gives 8 to his sister. How many stickers does Ali have now? 25 + 10 = 35, 35 - 8 = 27.

Watch for: A story where the second step starts from 25 instead of 35.; No question at the end.

G3. Mei, Ali and Ben have 70 shells altogether. Mei has 25 shells. Ali has 10 more shells than Mei. Write one number sentence for each step. How many shells does Ben have?

Answer: Ali has 35. Mei and Ali have 60. Ben has 10 shells.

Step 1: Ali has 25 + 10 = 35. Step 2: Mei and Ali have 25 + 35 = 60. Step 3: Ben has the rest: 70 - 60 = 10 shells.

Watch for: Answering 35 or 60 (stopping early).; Taking only Mei's shells from 70.

Investigation (page 5): Does the order matter?

Both orders give the same answer (20 + 10 − 5 = 25 and 20 − 5 + 10 = 25) whenever both orders can be done. But the order can matter for what you are allowed to do: from 4 you cannot take away 6 first, though 4 + 10 − 6 = 8 works. So: the answer is the same, but you cannot take away more than you have.

Look for: A child who tests several numbers, says what they notice, and finds a case where taking away first is not possible (start 4, take away 6).

Exit ticket (page 5)

E1. A bus has 25 children. 10 children get off. Then 3 children get on. How many children are on the bus now?

Answer: 18 (25 − 10 = 15, then 15 + 3 = 18)

Get off means take away: 25 − 10 = 15. Get on means add: 15 + 3 = 18.

Watch for: Writing 38 (adding the 10).; Writing 15 (stopping after Step 1).

E2. Ben has 6 stickers. Mei has 5 more stickers than Ben. How many stickers does Mei have?

Answer: 11 (6 + 5)

Mei has more than Ben, so add: 6 + 5 = 11.

Watch for: Writing 1 (taking away).; Writing 5.

E3. Siti has 20 beads. Ali has 7 fewer beads than Siti. How many beads does Ali have?

Answer: 13 (20 − 7)

Ali has fewer, so take away: 20 − 7 = 13.

Watch for: Writing 27 (adding because the number 7 is there).; Writing 7.

Sneak-peek puzzles (not in the workbook)

1. 7
2. 11. First find Ben's marbles: 4 + 3 = 7. Then add Ali's 4: 4 + 7 = 11.